

DANIEL J. DEUTSCH, M.Sc

Data Scientist & Machine Learning Engineer

Paris, FR | +33 7 45 10 02 91 | daniel.jrgde@gmail.com | danieljrgde | danieljrgde.github.io

Data Scientist with expertise in statistical modeling, causal inference, and production-grade machine learning in Python. Background spanning quantitative finance, NLP, and experimentation design across startup and enterprise environments, with a track record of translating complex analytical outputs into measurable business outcomes.

WORK EXPERIENCE

Machine Learning Engineer · CO2AI · Full-time · Paris, FR *Mar 2025 - Feb 2026*

- Automated complex entity resolution and mapping workflows leveraging a hybrid Transformer-fuzzy logic engine, increasing match accuracy by 20% and eliminating manual classification bottlenecks.
- Constructed weighted regression models for missing-data imputation across heterogeneous datasets, maximizing sample availability for downstream modeling while preserving statistical variance.
- Delivered production-grade Python modules within a 30+ engineer codebase, adhering to CI/CD pipelines, rigorous code review, and established software architecture standards.

Deep Learning Engineer · VO2 Finance · Full-time · Paris, FR *Sep 2023 - Sep 2024*

- Architected a multimodal Transformer network with a custom hybrid attention mechanism, integrating text and image tensors to achieve 96% AUC in high-dimensional compatibility prediction.
- Enhanced model generalization by constructing a proprietary dataset of 60k+ samples; utilized CLIP for zero-shot pseudo-labeling and LLaVA for synthetic captioning, reducing overfitting and improving validation accuracy by 5%.
- Implemented a RAG-based QA system using LangChain and Weaviate with hybrid search and prompt optimization, reducing hallucination rates by 35% and enabling reliable automated information retrieval.
- Presented the recommendation engine's architecture and business impact at VivaTech Paris 2024, translating multimodal AI concepts for executive and non-technical audiences.

Quantitative Research Analyst · Kaiko · Apprentice · Paris, FR *Sep 2021 - Sep 2022*

- Quantified the causal relationship between social signals and asset volatility by engineering a novel engagement-weighted sentiment metric derived from a BERT model fine-tuned on 6.6M+ posts to generate predictive features for alpha research.
- Investigated unsupervised anomaly detection methods (Isolation Forests, Autoencoders) on tick-level trading time-series, statistically flagging irregular patterns for market surveillance and compliance research.
- Processed terabytes of tick-level market data via PySpark on Hadoop, engineering statistical features for quantitative risk modeling and downstream analytical research pipelines.

Quantitative Data Engineer · BTG Pactual · Intern · São Paulo, BR *Dec 2019 - Aug 2020*

- Designed end-to-end ETL architectures for high-frequency trading data, normalizing multi-exchange datasets and optimizing ingestion workflows to support downstream quantitative research and ML feature generation.
- Implemented a serverless data validation framework for HFT ingestion pipelines (10 TBs/day), enforcing schema constraints, anomaly detection, and distribution-drift checks to protect trading and research systems from corrupted market data.
- Orchestrated dynamic EMR provisioning and distributed PySpark execution via Airflow, reducing compute overhead through automated cluster lifecycle management and fault-tolerant job design.

SKILLS

Technical Stack	Python (PyTorch, NumPy, Pandas, SciPy, Scikit-Learn, Statsmodels, PySpark, Transformers), Git, Docker, GCP, AWS.
Domain Expertise	NLP & Large Language Models, Time-Series Analysis, Anomaly Detection, Causal Inference, Automation.
Languages	English (fluent - C2), French (fluent - C1), Portuguese (native), Spanish (intermediate - B2).

PROJECTS

Bot Detection & Causal Analysis of Artificial Sentiment on Crypto Markets *Jul 2022*

- Isolated 13k bot accounts from a 162k user dataset by constructing a hybrid anomaly detection pipeline combining Variational Autoencoders and k-NN on behavioral feature embeddings.
- Executed Granger Causality tests on stationary crypto time-series to evaluate whether bot-driven signals held predictive power over asset volatility, statistically refuting common market manipulation hypotheses.

Algorithmic Fairness Audit in E-Commerce Search & Recommendation Systems *Apr 2021*

- Quantified algorithmic price steering across 10 distinct user personas by calculating Jaccard Similarity Indices and Kendall's τ coefficients, statistically proving that user browser history alters product search result rankings.
- Modeled Amazon's Buy-Box allocation using a Random Forest classifier achieving 80% accuracy, identifying price and seller rating as dominant drivers with implications for marketplace strategy and regulatory fairness assessments.

EDUCATION

Université Paris 1 Panthéon-Sorbonne | Master of Science in Quantitative Finance *Sep 2022*

Télécom Paris | Diplôme d'Ingénieur in Computer Science *Sep 2022*

Escola Politécnica da USP | Diplôme d'Ingénieur & Bachelor's in Electrical Engineering *Dec 2022*